

Assignment 1, Harmonic Analysis

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Problem 1

(a)

Since the half-diagonal of the cube $x + (-r/2, r/2)^d$ equals $r\sqrt{d}/2$, we have $x + (-r/2, r/2)^d \subseteq B(x, r\sqrt{d}/2)$. Setting $\rho = r\sqrt{d}/2$, so that $r^d = 2^d d^{-d/2} \rho^d$, gives

$$\begin{aligned} M'_c(f)(x) &= \sup_{r>0} \frac{1}{r^d} \int_{x+(-r/2, r/2)^d} |f(y)| \, dy \\ &\leq \sup_{r>0} \frac{1}{r^d} \int_{B(x, r\sqrt{d}/2)} |f(y)| \, dy \\ &= \sup_{\rho>0} \frac{d^{d/2}}{2^d \rho^d} \int_{B(x, \rho)} |f(y)| \, dy \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \sup_{\rho>0} \frac{1}{v_d \rho^d} \int_{B(x, \rho)} |f(y)| \, dy \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \cdot M_c(f)(x). \end{aligned}$$

(b)

Problem 2

Problem 3